

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\left. \begin{array}{l} \text{diky } \langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle \quad (\text{definice 8.14}) \\ \text{a diky } \langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle \quad (\text{poznámka 8.15}) \end{array} \right\} (*)$$

$$\begin{aligned} \Rightarrow \|u+v\|^2 - \|u-v\|^2 &= \langle u+v, u+v \rangle - \langle u-v, u-v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle - \langle v, v \rangle \\ &= 2\langle u, v \rangle + 2\langle v, u \rangle, \quad \text{protože platí } \langle u, v \rangle = \overline{\langle v, u \rangle} \\ &= 2\langle u, v \rangle + 2\overline{\langle u, v \rangle} \quad \text{z definice skal. součinu} \\ &= 2(\langle u, v \rangle + \overline{\langle u, v \rangle}) \\ &= 2(2\operatorname{Re}(\langle u, v \rangle)) \quad \text{protože } x + \bar{x} = 2\operatorname{Re}(x) \\ &= 4\operatorname{Re}(\langle u, v \rangle) \end{aligned}$$

$$\Rightarrow \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) = \operatorname{Re}(\langle u, v \rangle)$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$\text{diky } (*) \Rightarrow$$

$$\begin{aligned} \|u-iv\|^2 - \|u+iv\|^2 &= \langle u-iv, u-iv \rangle - \langle u+iv, u+iv \rangle \\ &= \langle u, u-iv \rangle + \langle -iv, u-iv \rangle - (\langle u, u+iv \rangle + \langle iv, u+iv \rangle) \\ &= \langle u, u \rangle + \langle u, -iv \rangle + \langle -iv, u \rangle + \langle -iv, -iv \rangle - (\langle u, u \rangle + \langle u, iv \rangle + \langle iv, u \rangle + \langle iv, iv \rangle) \end{aligned}$$

$$\text{Podle definice 8.14 a poznámky 8.15 } \langle tu, v \rangle = \bar{t} \langle u, v \rangle \quad \text{a} \quad \langle u, tv \rangle = t \langle u, v \rangle$$

$$\begin{aligned} &= \cancel{\langle u, u \rangle} - i\langle u, v \rangle + i\langle v, u \rangle - \underbrace{i\langle -iv, v \rangle}_{-i\bar{(-i)}\langle v, v \rangle} - \cancel{\langle u, u \rangle} - i\langle u, v \rangle + i\langle v, u \rangle - \underbrace{i\langle iv, v \rangle}_{-i\bar{i}\langle v, v \rangle} \\ &= -2i\langle u, v \rangle + 2i\langle v, u \rangle + \langle v, v \rangle - \langle v, v \rangle \end{aligned}$$

$$= -2i\langle u, v \rangle + 2i\overline{\langle u, v \rangle}$$

$$= -2i(\langle u, v \rangle - \overline{\langle u, v \rangle}) \quad \text{protože } z - \bar{z} = 2i\operatorname{Im}(z) \rightarrow \text{neboť } a+bi - (a-bi) = 2bi = 2\operatorname{Im}(z)$$

$$= -2i(2i\operatorname{Im}(\langle u, v \rangle)) = 4\operatorname{Im}(\langle u, v \rangle) \Rightarrow \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2) = \operatorname{Im}(\langle u, v \rangle)$$